

KIHYUN YU

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<https://kihyun-yu.github.io>

RESEARCH INTERESTS

My research focuses on **safe reinforcement learning** and sequential decision-making under constraints. I develop algorithms that learn to make effective decisions while respecting safety and resource constraints, with provable guarantees in challenging settings such as adversarial environments, function approximation, and infinite-horizon problems. I am also broadly interested in **bandits**, **online learning**, and their real-world applications.

EDUCATION

- KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST) South Korea
 - Integrated M.S.–Ph.D. Program in Industrial and Systems Engineering 03/2024 – present
 - Advisor: [Prof. Dabeen Lee](#)
 - GPA: 4.2 / 4.3
- GWANGJU INSTITUTE OF SCIENCE AND TECHNOLOGY (GIST) South Korea
 - B.S. in Electrical Engineering and Computer Science 03/2018 – 02/2024
 - Minor in Mathematics
 - GPA: 4.08 / 4.5 (*cum laude*)
 - Two-year leave for mandatory military service (2020 – 2021)
- SEJONG SCIENCE HIGH SCHOOL South Korea
 - 03/2015 – 02/2018

PUBLICATIONS

*: equal contribution

- [1] **Kihyun Yu**, Honghao Wei, Dabeen Lee. Rate-Optimal Algorithm for Adversarial Linear CMDPs. Available at: <https://arxiv.org/abs/2610.00927>
- [2] **Kihyun Yu***, Seoungbin Bae*, Dabeen Lee. Learning Infinite-Horizon Average-Reward CMDPs via State Augmentation. Available at: <https://arxiv.org/abs/2609.39093>
- [3] **Kihyun Yu**, Junehee Lee, Dabeen Lee. Constrained Online Convex Optimization without Slater's Condition. Available at: <https://arxiv.org/abs/2606.31480>
- [4] **Kihyun Yu**, Beomhan Baek, Dabeen Lee. Learning Weakly Communicating Average-Reward CMDPs: Strong Duality and Improved Regret. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Available at: <https://arxiv.org/abs/2605.11586>
- [5] **Kihyun Yu**, Seoungbin Bae, Dabeen Lee. Near-Optimal Primal-Dual Algorithm for Learning Linear Mixture CMDPs with Adversarial Rewards. *IEEE Conference on Decision and Control (CDC)*, 2026. Available at: <https://arxiv.org/abs/2603.27884>
- [6] **Kihyun Yu**, Seoungbin Bae, Dabeen Lee. Primal-Dual Policy Optimization for Linear CMDPs with Adversarial Losses. *International Conference on Learning Representations (ICLR)*, 2026. Available at: <https://arxiv.org/abs/2605.11535>
- [7] Jiahui Zhu, **Kihyun Yu**, Dabeen Lee, Xin Liu, Honghao Wei. An Optimistic Algorithm for online CMDPs with Anytime Adversarial Constraints. *International Conference on Machine Learning (ICML)*, 2025. Available at: <https://arxiv.org/abs/2505.21841>

- [8] **Kihyun Yu**, Duksang Lee, William Overman, and Dabeen Lee. Improved Regret Bound for Safe Reinforcement Learning via Tighter Cost Pessimism and Reward Optimism. *Reinforcement Learning Conference (RLC)*, 2025. Available at: <https://rlj.cs.umass.edu/2025/papers/Paper41.html>

ACADEMIC SERVICES

- Conference Reviewer: ICML 2026 ([silver reviewer](#)), NeurIPS 2026, IEEE CDC 2026, AAAI 2027, ICLR 2027
- Journal Reviewer: TMLR, IEEE Transactions on Networking

TEACHING EXPERIENCE

- | | |
|---|-------------------|
| • Teaching Assistant, Operations Research: Optimization (KAIST) | 03/2024 – 06/2024 |
| • Teaching Assistant, Differential Equations (GIST) | 09/2023 – 12/2023 |
| • Teaching Assistant, Single Variable Calculus (GIST) | 03/2023 – 06/2023 |

HONORS

- Scholarship for Academic Excellence
 - Awarded for 6 semesters (Spring 2018, Fall 2018, Spring 2019, Fall 2019, Spring 2022, Spring 2023)

EXPERIENCE

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| • Information Processing, Controlling, and Network Lab (GIST) | South Korea |
| - Undergraduate Research Intern (Advisor: Prof. Heung-No Lee) | 01/2023 – 12/2023 |
| - Conducted research on lattice-based cryptography | |
| • Artificial Intelligence Lab (GIST) | South Korea |
| - Undergraduate Research Intern (Advisor: Prof. Kyoobin Lee) | 07/2022 – 12/2022 |
| - Conducted research on robust image classification under noisy labels | |
| • University of Cambridge | United Kingdom |
| - Summer Session Programme (Machine Learning) | 07/2019 – 08/2019 |

LANGUAGES AND TECHNICAL SKILLS

- Korean (native), English (conversational)
- C/C++, Python, PyTorch, $\text{\LaTeX}$

OTHER ACTIVITIES

- Volunteer Math Mentor 03/2018 – 12/2018
 - Provided small-group math mentoring for middle and high school students